
Back to entropy

Evaluating language models as language models

David Afonso Valente · Bottlecap AI

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How good is this model, for what we do?

A new base model ships most weeks. What arrives with it is a scorecard.

- A language model is trained to do one thing: **predict text**.
- So the direct question is: how well does it predict *our* text?
- Instead we read MMLU-Pro, HellaSwag, GSM8K.

Llama-3.2-1B vs Qwen2.5-1.5B

HellaSwag	2.1 points apart
Adapted BPB on news	3.4% apart
GSM8K	55.3 points apart

Two models that model fresh English almost identically well differ by nearly a factor of ten on one exam. Which number describes the model?

Two reasons the scorecard misleads

1. Benchmarks are public

- Written before the model existed, for other purposes.
- Vendors optimise for the ones they report.
- Items and subject matter can reach pre-training data.
- Stronger model, or saw more of the exam?
No reader can tell.

2. They score a model before it is adapted

- Teams adapt a base model to their data first.
- What matters is how good it *becomes*.
- **Gemma-4-12B**, of fifteen models: 9th on GSM8K, 8th on MMLU-Pro, 13th on raw loss ...
- ... and **the best fine-tuned system of all**.

Why loss is the right quantity

There is no universal ground truth for “better” — so pick the principled yardstick

- Every base model was trained to minimise cross-entropy on text. Held-out loss is **its own objective**, measured out of sample.
- Cross-entropy is a proper scoring rule. In expectation over the domain's text p :

$$H(p, q) = \underbrace{H(p)}_{\text{same for every model}} + \text{KL}(p \parallel q)$$

so a difference in loss *is* a difference in distance from the text's distribution.

- The one obstacle: each model counts in its own tokens. Divide by **bytes**.

Bits per byte is not one metric among many. It is the training objective, put on a common scale.

Taking that seriously: three requirements

Bytes

every tokenizer
on one scale

Fresh, decontaminated text

no candidate
has trained on it

Equal adaptation

the same procedure
for every candidate

Decontamination, three layers:

- **Time** — text dated after every model shipped (Reddit and Hacker News, most of arXiv).
- **Hygiene** — exact dedup, quality filter, MinHash near-dup removal, 80/10/10 split with no overlap.
- **Membership screens** — Min-K++, context sensitivity, perturbation; the union of anything any model flags is quarantined for all (news snapshot).

Equal adaptation: the procedure, and what we ran

The method

-hparams sweep, default

- Per-model Optuna search: TPE sampler over LR, batch, warm-up, weight decay, LoRA rank, α , dropout.
- Successive halving prunes weak trials from 50 steps.
- Scored on the *validation* split; test never touched.
- Same number of trials for every model — no time cap.
- Winner trained until validation BPB stops improving (-no-time-limit).

What these results use

-hparams manual

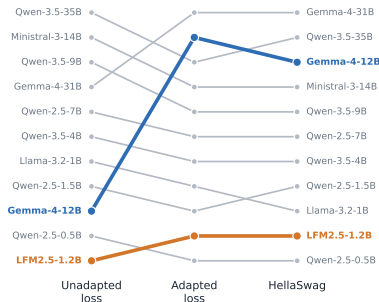
- A full search of 17 models up to 35B was beyond our compute.
- One hand-picked recipe: LoRA $r=16$, $\alpha=32$, lr 10^{-4} , batch 32, fixed 250 steps.
- One A100: **3.7 min** at 0.5B, 11.6 at 7.7B, 24.7 at 12B, 62.7 at 31B.
- Where our earlier search settled: rank 16 won for 8 of 11 models.

A time-capped search is a trap: small models finish more trials and get better tuned. In our first run the 1B models completed 17–18 trials, the two largest only 4.

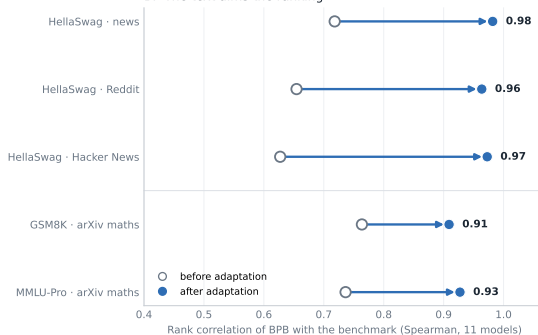
Adaptation reorders the models — and lands on the language benchmark

Left: eleven models ranked three ways on news. Right: what the adapted ordering tracks.

A. Eleven models on news, ranked three ways
reorders *converges*



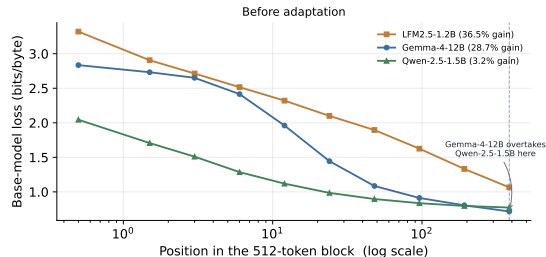
B. The text aims the ranking



Gemma-4-12B climbs from ninth to second; after adaptation the order is HellaSwag's (Spearman **0.96–0.98**). Adapted on maths instead, it swings to GSM8K and MMLU-Pro.

1. Adaptation reorders the models

Eleven models; unadapted vs adapted rankings agree only at $\rho = 0.61$ – 0.68



- Adaptation cuts loss by **3.6–5.4%** for every Qwen, Llama and Ministral model
...
- ... but by **17–36%** for the two Gemmas and LFM — the *only* models that climb.
- Gemma-4-12B: 9th → 2nd. Unadapted, it opens each block at 2.83 BPB vs a 1.5B model's 2.04, and passes it by the end.

Unadapted loss mixes *familiarity with this prose* into capability. A short equal adaptation removes the familiarity.

2. It converges on the language benchmark — and can be aimed

Rank correlation of BPB with each benchmark, before and after adaptation (eleven models)

	Before adaptation	After adaptation
HellaSwag, on news / Reddit / Hacker News	0.63–0.72	0.96–0.98
GSM8K, adapted on arXiv mathematics	0.76	0.91
MMLU-Pro, adapted on arXiv mathematics	0.74	0.93

- HellaSwag is the benchmark closest to language modelling — and adapted BPB lands on it.
- Change the text to mathematics and the ranking swings to the reasoning benchmarks.

3. It is stable where benchmarks are not

Adapted BPB

- Seed-to-seed SD is **0.0012**× the spread between models; 0 of 40 close pairs flip.
- Re-run weeks later on other hardware: within 0.0003.
- Adapted ranking across news / Reddit / Hacker News: $\rho = 0.955\text{--}0.991$.

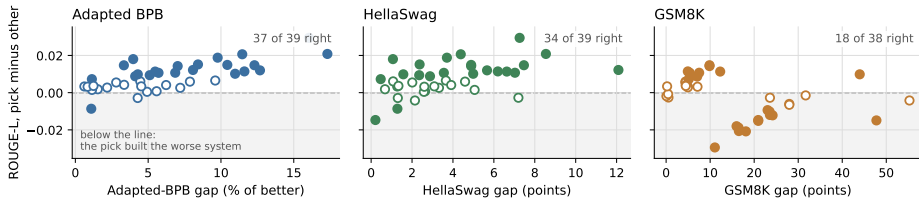
Generative benchmarks

- Same weights, new host (A100 → H100): GSM8K moves **−0.60 to +0.69** points.
- One shipped decoding setting: **9.86** GSM8K and **14.30** MMLU-Pro points for Gemma-4-12B.

A generative benchmark measures the model, the harness, the decoding and the hardware at once. Bits per byte on a fixed corpus measures the model.

4. Minutes of adaptation anticipate hours of fine-tuning

Fifteen models fine-tuned on news; 39 size-matched pairs



Adapted BPB, measured *before* any task existed, orders **0.949** of pairs as the fine-tune does — HellaSwag 0.872, MMLU-Pro 0.513, GSM8K 0.474. **Every model within one rank** of where it finishes. Holds under a second, very different recipe (0.923).

Using it — and adding your model

- Candidates more than $\sim 2\times$ apart in size? The larger is usually better.
- Otherwise: recent domain text, decontaminated; adapt every finalist the same way; rank by adapted BPB.
- A gap under **2%** is a tie — let licence, latency or cost decide.
- The zero-shot number is a free screen, not the choice.

Public leaderboard, two tracks: [arXiv maths](#) (CC0 corpus in the repo — anyone can run it and submit) and [news](#) (maintainers run requested models). Submissions are checked against the track's corpus hash and recipe.

Evaluate language models as language models.

Questions?

Backup: per-model BPB, benchmark scores, contamination breakdown, longer-context transfer.

Backup: the adaptation results as a table

Lower BPB is better. Hand-picked recipe, 250 steps, news corpus.

Model	Architecture	Params	Zero-shot	Adapted	Reduction
Gemma-4-31B	Dense Transformer	31B	0.752	0.624	17.0%
Gemma-4-12B	Dense Transformer	12B	0.866	0.653	24.5%
Qwen-3.5-35B-MoE	Mixture of Experts	35B	0.707	0.678	4.1%
Ministral-3-14B	Dense Transformer	14B	0.718	0.679	5.4%
Qwen-3.5-9B	Dense Transformer	9B	0.740	0.708	4.4%
Qwen-2.5-7B	Dense Transformer	7B	0.758	0.717	5.4%
Qwen-3.5-4B	Dense Transformer	4B	0.782	0.749	4.2%
Llama-3.2-1B	Dense Transformer	1.2B	0.811	0.771	4.9%
Qwen-2.5-1.5B	Dense Transformer	1.5B	0.833	0.797	4.3%
LiquidAI-LFM2.5	Liquid Neural Net	1.2B	1.340	0.852	36.4%
Qwen-2.5-0.5B	Dense Transformer	0.5B	0.935	0.901	3.6%

Every value reproduces from the marker-masked fixed-protocol files in `results/domain_transfer/news_*.json`.

Backup: full benchmark scores

Eleven-model cohort, one host, one build

Model	Adapted BPB	MMLU-Pro	HellaSwag	GSM8K
Gemma-4-31B	0.624 (1)	57.1 (2)	87.9 (1)	85.0 (3)
Gemma-4-12B	0.653 (2)	44.8 (7)	84.0 (3)	64.8 (7)
Qwen-3.5-35B-MoE	0.678 (3)	61.4 (1)	85.6 (2)	86.6 (1)
Ministral-3-14B	0.679 (4)	52.8 (4)	82.9 (4)	80.8 (6)
Qwen-3.5-9B	0.708 (5)	55.6 (3)	81.6 (5)	85.8 (2)
Qwen-2.5-7B	0.717 (6)	47.2 (6)	80.3 (6)	81.3 (4)
Qwen-3.5-4B	0.749 (7)	52.4 (5)	77.2 (7)	81.0 (5)
Llama-3.2-1B	0.771 (8)	11.2 (11)	65.8 (9)	6.4 (11)
Qwen-2.5-1.5B	0.797 (9)	25.6 (8)	68.0 (8)	61.6 (8)
LiquidAI-LFM2.5	0.852 (10)	16.4 (9)	60.9 (10)	54.1 (9)
Qwen-2.5-0.5B	0.901 (11)	16.0 (10)	51.8 (11)	34.3 (10)

Percent, rank in parentheses; generated from the committed harness logs, the same table as the paper's. *lm-evaluation-harness* 0.4.12, bf16, greedy, native BOS; MMLU-Pro 5-shot CoT on a fixed 1,000-question subset, HellaSwag 10-shot length-normalised, GSM8K 5-shot flexible-extract.

Backup: contamination breakdown

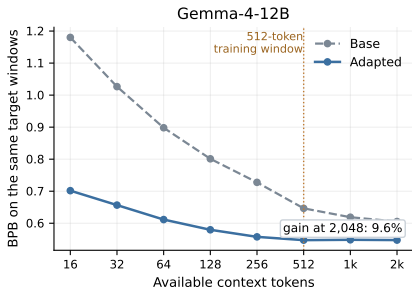
$N = 1000$ sampled sequences per model; 80 unique risk-flagged sequences removed

Model	Min-K++ flags	CoDeC flags	DCQ candidates	Preference	Risk flags
google-gemma-4-31B	12	6	20	100%	20
google-gemma-4-12B	11	6	11	100%	11
mistralai-Ministral-3-14B	12	8	13	100%	13
Qwen-Qwen3.5-35B-MoE	6	8	11	100%	11
Qwen-Qwen3.5-9B	12	6	17	100%	17
Qwen-Qwen2.5-7B	10	6	14	100%	14
Qwen-Qwen3.5-4B	9	6	12	100%	12
meta-llama-Llama-3.2-1B	17	0	28	100%	28
Qwen-Qwen2.5-1.5B	14	4	17	100%	17
LiquidAI-LFM2.5	12	8	23	96%	22
Qwen-Qwen2.5-0.5B	8	2	13	100%	13

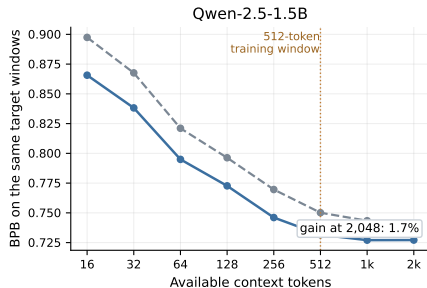
These are conservative quarantine signals on a preselected subset, not verified memorization rates. Exact overlap and model-likelihood signals must be interpreted separately.

Backup: 512-token adaptation transfers to longer contexts

Same held-out targets; only the available preceding context changes



Gemma base BPB keeps recovering after 512:
0.6465 → **0.6054** at 2,048 tokens.



Adapted Gemma stays at **0.547**; its advantage remains **9.6%** at four times the training context.

The adapter does not require the inference sequence to stop at 512 tokens.